A SEPARATED, LOCALLY QUASI-FINITE MORPHISM OF ALGEBRAIC SPACES TO A REPRESENTABLE BASE HAS REPRESENTABLE SOURCE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's reduction-to-schemes argument and ends with the sentence that the proof is finished; the conclusion is not left to the reader.

Editorial note. The fifteen references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Representability of an algebraic space is Zariski local on the space and on the base. Properties of Spaces, Lemma spaces-properties-lemma-subscheme, used at proof lines 10822-10823, 10837-10838 and 10910. Reduces the claim to a covering family of open subspaces and closes the argument.
- Stability of separatedness and local quasi-finiteness under base change. Lemmas lemma-base-change-separated and lemma-base-change-quasi-finite, used at proof lines 10824-10828. Allows replacing T by an affine open and X by its base change.
- Etale equivalence relations and the presentation X = U/R by an affine scheme U. Spaces, Lemma spaces-lemma-space-presentation, used at proof lines 10854-10855. Produces the groupoid to which the More on Groupoids lemma is applied.
- The quasi-finiteness lemma for groupoids over a base with a chosen pair of points. More on Groupoids, Lemma more-groupoids-lemma-quasi-finite-over-base-j-proper, used at proof lines 10870-10886. Supplies the etale neighbourhood and the disjoint union decompositions of U and R.
- Finite flat equivalence relations have affine quotient. Groupoids, Proposition groupoids-proposition-finite-flat-equivalence, used at proof lines 10895-10897. Makes U'/R' an affine scheme, hence an open subscheme of X_{T'}.

Source reference. Original proposition 03BM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4353–4353. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03XL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5152–5152. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4681–4681. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03XK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5151–5151. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03XU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5161–5161. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4683–4683. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03XW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5163–5163. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03X5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5136–5136. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02GS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3239–3239. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02GR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3238–3238. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03WS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5123–5123. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3817–3817. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0262: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2835–2835. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03JH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4639–4639. Navigation pointer into the Stacks

Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03FY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4509–4509. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Proposition 0.2. *Let S be a scheme. Let $f : X \to T$ be a morphism of algebraic spaces over S. Assume*

(1) *T is representable,*
(2) *f is locally quasi-finite, and*
(3) *f is separated.*

Then X is representable.

Proof. Let $T = \bigcup T_i$ be an affine open covering of the scheme T. If we can show that the open subspaces $X_i = f^{-1}(T_i)$ are representable, then X is representable, see Properties of Spaces, Lemma 03JH. Note that $X_i = T_i \times_T X$ and that locally quasi-finite and separated are both stable under base change, see Lemmas 03KL and 03XL. Hence we may assume T is an affine scheme.

By Properties of Spaces, Lemma 03FY there exists a Zariski covering $X = \bigcup X_i$ such that each X_i has a surjective étale covering by an affine scheme. By Properties of Spaces, Lemma 03JH again it suffices to prove the proposition for each X_i. Hence we may assume there exists an affine scheme U and a surjective étale morphism $U \to X$. This reduces us to the situation in the next paragraph.

Assume we have

$$U \longrightarrow X \longrightarrow T$$

where U and T are affine schemes, $U \to X$ is étale surjective, and $X \to T$ is separated and locally quasi-finite. By Lemmas 03XU and 03XK the morphism $U \to T$ is locally quasi-finite. Since U and T are affine it is quasi-finite. Set $R = U \times_X U$. Then $X = U/R$, see Spaces, Lemma 0262. As $X \to T$ is separated the morphism $R \to U \times_T U$ is a closed immersion, see Lemma 03KN. In particular R is an affine scheme also. As $U \to X$ is étale the projection morphisms $t, s : R \to U$ are étale as well. In particular s and t are quasi-finite, flat and of finite presentation (see Morphisms, Lemmas 03WS, 02GS and 02GR).

Let (U, R, s, t, c) be the groupoid associated to the étale equivalence relation R on U. Let $u \in U$ be a point, and denote $p \in T$ its image. We are going to use More on Groupoids, Lemma 03X5 for the groupoid (U, R, s, t, c) over the scheme T with points p and u as above. By the discussion in the previous paragraph all the assumptions (1) – (7) of that lemma are satisfied. Hence we get an étale neighbourhood $(T', p') \to (T, p)$ and disjoint union decompositions

$$U_{T'} = U' \amalg W, \quad R_{T'} = R' \amalg W'$$

and $u' \in U'$ satisfying conclusions (a), (b), (c), (d), (e), (f), (g), and (h) of the aforementioned More on Groupoids, Lemma 03X5. We may and do assume that T' is affine (after possibly shrinking T'). Conclusion (h) implies that $R' = U' \times_{X_{T'}} U'$ with projection mappings identified with the restrictions of s' and t'. Thus $(U', R', s'|_{R'}, t'|_{R'}, c'|_{R' \times_{t', U', s'} R'})$ of conclusion (g) is an étale equivalence relation. By Spaces, Lemma 02WU we conclude that U'/R' is an open subspace of $X_{T'}$. By conclusion (d) the schemes U', R' are affine and the morphisms $s'|_{R'}, t'|_{R'}$ are finite étale. Hence Groupoids, Proposition 03BM kicks in and we see that U'/R' is an affine scheme.

We conclude that for every pair of points (u, p) as above we can find an étale neighbourhood $(T', p') \to (T, p)$ with $\kappa(p) = \kappa(p')$ and a point $u' \in U_{T'}$ mapping to u such that the image x' of u' in $|X_{T'}|$ has an open neighbourhood V' in $X_{T'}$ which is an affine scheme. We apply Lemma 03XW to obtain an open subspace $W \subset X$ which is a scheme, and which contains x (the image of u in $|X|$). Since this works for every x we see that X is a scheme by Properties of Spaces, Lemma 03JH. This ends the proof. $\square$